

- (c) It will rotate into the orbit of Earth
(d) It will burst out

R.A.S./R.T.S. (Pre) 1992

Ans. (b)

11.2 km/sec is the escape velocity of the Earth which is minimum speed that is required to escape from the Earth's gravity. If an object thrown with the speed less than this, it will return to Earth.

39. There is no atmosphere on the Moon because –

- (a) It is near to the Earth.
(b) It receives light from the Sun.
(c) It revolves around the Earth.
(d) The escape velocity of gas molecules on it is less than the root mean square velocity here.

U.P.P.C.S. (Mains) 2012

Ans. (d)

Moon has no atmosphere because the value of acceleration due to gravity 'g' on the surface of the Moon is small. Therefore, the value of escape velocity on the surface of the Moon is small. The value of root mean square (RMS) velocity of the molecules of different gases is much above the value of escape velocity on the Moon. That is why all the molecules of gases escaped and there is no atmosphere on the Moon.

40. There is no atmosphere on the surface of the moon because :

- (a) It is closer to Earth
(b) Escape velocity of gas molecules is less than RMS velocity
(c) It revolves around Earth
(d) It gets light from Sun

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (b)

See the explanation of above question.

41. Given below are two statements–

Assertion (A) : Lighting a candle in space does not produce a flame.

Reason (R) : A flame exists due to the pull of gravity. In the context of above, which one of the following is correct?

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation.
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

U.P.P.C.S. (Pre) 2009

Ans. (a)

NASA made it clear by his experiment that gravitation is must for lighting up a candle. In the case of microgravity in the spacecraft, we are able to light up the flame but it is not possible in an actual vacuum. In fact, the flickering frequency of a flame is proportional to the square root of the acceleration due to gravity to the diameter of the candle and due to the lack of gravitational force in space, it is zero. Thus, the correct Answer is (a).

42. In the state of weightlessness, the size of candle's flame will be –

- (a) Longer (b) Smaller
(c) Spherical (d) Constant

R.A.S./R.T.S. (Pre) 2005

Ans. (c)

In the state of weightlessness or microgravity, the size of candle's flame will be spherical in shape.

Physical Properties of Matter

Notes

- Generally, there are three states of matter - (i) Solid, (ii) Liquid, and (iii) Gas.
- The three states of matter have some special properties, which are as follows -
 - In solid - Elasticity.
 - In liquid - Pressure, Surface Tension, Capillarity and Floating.
 - In Gas - Atmospheric Pressure.

Elasticity :

- Generally, the meaning of a rigid substance is a hard solid which has definite shape and size.
- But actually these can be stretched, compressed and deformed.
- Even though a rigid steel rod can be deformed by applying sufficient external force.
- When an external force is applied on any substance, its both shape or size are deformed.
- After withdrawing force, it gains its normal shape or size.
- The property of any substance by which it tries to gain its normal shape after being stretched or compressed is called elasticity.
"Ability of an object or material to resume its normal shape after being stretched or compressed" is elasticity.
- On the basis of the property of elasticity, matters are of two types :
 - Perfectly elastic substance** - Matter which resumes perfectly its normal state after withdrawing the applied force.
 - Perfectly plastic substance** - Matter which does not

resume its normal state after withdrawing the external force, but deformed forever.

- Really no matter is perfectly elastic nor perfectly plastic but lies between the limit of these two properties.
- On whole quartz can be considered as perfectly elastic matter and wax is perfectly plastic matter.

Stress :

- During the application of external force on a matter to change its shape or size, an internal reaction force of equal amount is produced in the opposite direction on every section.
- This force tries to resume a normal state of matter and it is known as stress.
- If force (F) is applied vertically on a transverse section with the area (A) of matter, then

$$\text{Stress} = F/A$$
 Its SI unit is newton/metre².

Strain :

- When the shape or size of a matter is changed by applying an external force, it is known as strain. Since, it is a ratio, hence it is without any unit.

Hooke's Law –

- If the strain in the matter is nominal, applied stress on the matter is directly proportional to strain developed in it. The ratio of stress and strain is a constant value, known as **elasticity coefficient (E)**.

$$E = \frac{\text{stress}}{\text{strain}}$$

Surface Tension :

- Surface Tension is the elastic tendency of a fluid surface which makes it to acquire the least surface area possible. It is measured in force per unit length.
- Its SI unit is - newton/metre.
- The value of surface tension of any liquid depends upon its nature, temperature and the medium present on another side of the liquid surface.
- Surface tension decreases with rise in temperature.
- It becomes zero at the critical temperature.
- If the liquid has negligible weight, its shape will be completely spherical. For example - rainwater drops and soap bubbles.
- The reason is drops occupy that form, in which its potential energy is minimum.
- It only surface tension is acting on drop, its potential energy will be minimum with less area. The rain- water drop will become spherical.

Cohesive and Adhesive Force :

- According to molecular theory, every matter is made up of small particles, known as molecules. These molecules attract each other.

- Cohesive force is the force of attraction between molecules of the same matter.
- Adhesive force is the force of attraction between molecules of different matter.
- The adhesive force between oil & water is less than the cohesive force of water but higher than cohesive force of oil, so when the water dropped on oil surface, its drops occupy spherical shape by shrinking, while when oil dropped on the water surface, it spreads widely in form of the thin film.

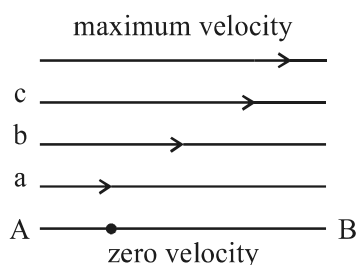
Capillarity :

- The capillary tube is a glass tube with a very small diameter, opened at both ends.
- The tendency of a liquid to rise or fall as a result of surface tension in the capillary tube is known as capillarity.
- Liquids which wet the glass, rise up and liquids which does not wet the glass, fall down in the capillary tube.
- For example – when the capillary tube is kept stand in mercury, the mercury surface falls down in tube in comparison to the surface of external mercury.

Examples of capillarity –

- Water and minerals absorbed by plant roots reached to leaves and twigs through xylem tissues - which act as capillary tubes.
- Drawing of ink to the nib from bottom in a fountain pen. The nib is slitted in the middle, which acts as a capillary tube. However, refill pen writing is possible by gravity and it does not depend upon surface tension.
- Lighting candles and lamp wicks.
- Blotting paper used to absorb spilled liquid.

Viscosity :



- A liquid is flowing in a rigid horizontal surface AB as shown in the figure. The liquid layer which is in contact with the bottom remains constant but the velocity of the upper layers are increasing. The uppermost layer flows with maximum velocity.
- In three horizontal layers a, b and c, a is with minimum velocity, b has more than a and c is with maximum velocity.
- The consequent layers, from bottom to surface try to slow down the velocity of the layer which is just above it.

- An internal force acts on different layers, try to slow down the velocity of the layer which is just above it.
- An internal force acts on different layers which tries to oppose their relative velocity.
- This force is known as the viscous force.
- To maintain the flow of liquid, the external force is applied against viscous force.
- The property of liquid which opposes the relative movement among different layers of liquid is known as viscosity.
- Concentrated liquids like honey and glycerine have greater viscosity, so their flow stops rapidly, than a low viscosity fluid.
- When temperature increases, the viscosity of liquid decreases.
- The viscosity of gas increases by increasing in temperature.
- Solids lack viscosity.

Pressure :

- The pressure is a force applied perpendicularly on per unit area surface of an object.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

SI unit of Pressure is newton/metre² (Pascal).

Pressure within liquid at any point –

- Pressure at a point, situated at h depth from the free surface of the liquid is –

$$P = h \times d \times g$$

where, d is density of liquid
g is gravitational acceleration.

Upthrust Force :

- When a solid is put into the liquid, loss in weight of object appears. This apparent loss is due to a force which is exerted by the liquid on the matter in the upper direction. This force is known as the upthrust force.
- The centre of upthrust of a matter is the centroid of the displaced volume of liquid.

Archimedes' Principle :

- It states that the upward force that is exerted on a body immersed in a liquid, whether fully or partially submerged, is equal to the weight of the liquid that the body displaces and it acts in the upward direction at the centre of mass of the displaced liquid.

Law of Floatation :

- When a solid is in liquid, the following two forces act on it-
1. The weight of solid acts perpendicularly inwards.
2. Upthrust force of liquid acting upwards on solid, in perpendicular direction.
- The value of F (upthrust) is equal to the weight of the liquid displaced by the solid.
- The sinking or floating of any object in liquid depends upon the relative values of these two forces. conditions are as follows –

- (i) If $W > F$ – resulting force ($W - F$) will act inwards–object will sink.
- (ii) If $W = F$ – object will float.
- (iii) If $W < F$ – leaving object free, it will start to come upward.
- How much portion of solid will remain in liquid, it depends upon the relative density of liquid and solid which can be calculated as follows –

$$\frac{\text{Volume of sinked solid portion}}{\text{Total volume of solid}} = \frac{\text{Density of solid}}{\text{Density of liquid}}$$
- From the above equation, it is clear that solid will sink less in liquid with high density and will sink more in liquid with less density.
- If the density of both solid and liquid are equal, solid will float.
- The liquid which is having less density will float on the liquid having more density.

Question Bank

1. A form of matter has no fixed shape but it has a fixed volume. An example of this form of matter is :

- (a) Carbon steel
- (b) Krypton
- (c) Kerosene
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Among the given options, kerosene is liquid, which has no fixed shape but it has a fixed volume. Carbon steel is a solid and has definite shape and volume, while krypton is a gas which has neither a definite shape nor volume.

2. With reference to the mechanical properties of solids, which of the following statement/s is/are correct?

- (1) The Young's Modulus of rubber is greater than that of steel.
- (2) The stretching of a coil is determined by its Shear Modulus.

- (a) Only (2)
- (b) Neither (1) nor (2)
- (c) Both (1) and (2)
- (d) Only (1)

U.P. R.O./A.R.O. (Pre) 2023

Ans. (a)

The Young's modulus (E), also known as modulus of elasticity, is a property of the material that tells us how easily it can stretch and deform and is defined as the ratio of tensile stress to tensile strain. Where stress is the amount of force applied per unit area and strain is extension per unit length. The Young's modulus for steel is much more than that for rubber. For same longitudinal strain, steel will have greater tensile stress than that of rubber. Thus, statement (1) is incorrect.

The shear modulus, also known as the modulus of rigidity, is a material property that measures the stiffness of a solid material in response to shear stress. It describes a material's ability to resist deformation when subjected to forces acting parallel or tangentially to its surface. It is the measure of the rigidity of the body, given by the ratio of shear stress to shear strain. The stretching of a coil changes its shape. Hence, it is determined by its shear modulus. Thus, statement (2) is correct.

3. Spherical form of raindrop is due to –

- (a) Density of liquid (b) Surface tension
(c) Atmospheric pressure (d) Gravitational force

Chhattisgarh P.C.S. (Pre) 2003

39th B.P.S.C. (Pre) 1994

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

The shape of a drop of rain is constrained by the surface tension, which tries to give it the shape for which the surface area is minimum for the given volume. The spherical shape has the minimum surface area. That's why rain drops acquire spherical shape.

4. A drop of water is spherical in shape. This is due to :

- (a) Surface tension (b) Low temperature
(c) Air resistance (d) Viscosity of water

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

See the explanation of above question.

5. The rain drops are spherical because –

- (a) They fall from big height
(b) The air has resistance
(c) The water has surface tension
(d) None of the above

U.P.P.C.S. (Pre) 2005

Ans. (c)

See the explanation of above question.

6. The tendency of a liquid drop to contract and occupy minimum area is due to :

- (a) Surface tension (b) Viscosity
(c) Density (d) Vapour pressure

I.A.S. (Pre) 1997

Ans. (a)

See the explanation of above question.

7. Oil spreads on water surface because –

- (a) Oil is denser than water
(b) Oil is less dense than water

(c) Surface tension of oil is more than water

(d) Surface tension of oil is less than water

U.P.P.S.C. (GIC) 2010

U.P.P.C.S. (Pre) 1994

Ans. (d)

Oil spreads on water surface because the surface tension of oil is less than water. Therefore, when oil is poured over water, the greater value of surface tension of water pulls oil in all directions, and as such it spreads on the water.

8. Kerosene oil floats on water because –

- (a) Its density is greater than the density of water.
(b) Its density is less than the density of water.
(c) Its density is equal to that of the density of water.
(d) None of the above

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

Kerosene oil floats on water because its density is less than the density of water.

9. Pressure inside a soap bubble is –

- (a) More than atmospheric pressure
(b) Less than atmospheric pressure
(c) Equal to atmospheric pressure
(d) Half of atmospheric pressure

U.P.P.C.S. (Pre) 2014

U.P.P.C.S. (Pre) 1995

Ans. (a)

Soap bubbles are large because when soap dissolved in water its surface tension is reduced. The pressure inside a soap bubble is more than atmospheric pressure.

10. If two bubbles of soap of different diameter come in contact with a tube, then what will happen?

- (a) The size of both the bubbles will remain same
(b) The smaller bubble will become smaller and bigger will become bigger
(c) The smaller bubble will become larger and larger will become smaller to get the same size
(d) Both bubbles will burst as soon as they come in contact

R.A.S./R.T.S.(Pre) 2007

Ans. (b)

The excess pressure inside the soap bubble is inversely proportional to the radius of soap bubble i.e. $p \propto \frac{1}{r}$. When these two bubbles are connected by a tube, the air will flow from smaller bubble to bigger bubble due to the fact that small bubble having smaller radius will exert more pressure. Hence the bigger bubble would grow at the expense of the smaller one.

11. A hydrogen-inflated polythene balloon is released from the surface of the Earth. As the balloon rises to an altitude up in the atmosphere, it will –

- (a) Decrease in size
- (b) Flatten into disc-like shape
- (c) Increase in size
- (d) Maintain the same size and shape

I.A.S. (Pre) 2003

Ans. (c)

If a hydrogen-inflated polythene balloon is released from the surface of the Earth, then the outside air pressure at high altitudes decreases while the pressure inside the balloon remains relatively constant. Thus, the balloon will expand until the pressure inside the balloon equals the pressure outside.

12. What happens when some charge is placed on a soap bubble?

- (a) Its radius increases
- (b) Its radius decreases
- (c) The bubble collapses
- (d) None of the above

U.P.P.C.S. (Pre) 2019

Ans. (a)

When some charge is placed on a soap bubble, the bubble will expand and its radius increases because the charge will distribute itself evenly over the surface of the bubble, and the repulsion between the same charges will cause it to expand until the surface tension can bring it to equilibrium. This will happen to both positive and negatively charged bubbles.

13. Four wires of same material and dimensions as mentioned below are stretched by a load of same magnitude separately. Which one of them will be elongated maximum?

- (a) Wire of 1 m length and 2 mm diameter
- (b) Wire of 2 m length and 2 mm diameter
- (c) Wire of 3m length and 1.5 mm diameter
- (d) Wire of 1m length and 1 mm diameter

I.A.S. (Pre) 2007

Ans. (c)

All the four wires are made up of the same material, therefore, they have the equal Young's modulus (modulus of elasticity)

$$Y = \frac{f/A}{\Delta l/L} = \frac{fL}{A\Delta l} = \frac{fL}{\pi r^2 \Delta l}$$

$$\therefore \Delta l \propto \frac{L}{r^2}$$

Thus, elongation of these wires is proportional to their length and inversely proportional to the square of radius/diameter. So option (c) has the maximum elongation.

14. Consider the following statement –

If there were no phenomenon of capillarity, then

- 1. It would be difficult to use a kerosene lamp.
- 2. One would not be able to use a straw to consume a soft drink.
- 3. The blotting paper would fail to function.
- 4. The big trees that we see around would not have grown on the Earth.

Which of the following statements given above are correct?

- (a) 1, 2 and 3 only
- (b) 1, 3 and 4 only
- (c) 2 and 4 only
- (d) 1, 2, 3 and 4 only

I.A.S. (Pre) 2012

Ans. (b)

Capillary action is the ability of a liquid to flow in narrow spaces without the assistance of, or even in opposition to external forces like gravity. Kerosene rises in the wick of the lamp through capillary action to use a straw. Blotting paper also works on the principle of capillary action. It has a large number of pores on its surface which acts like a capillary tube. Nature is an excellent user of capillary action. The fine roots of trees act as capillary tubes, which provide them useful nutrients and water. Hence, statements 1, 3 and 4 are correct. If a person wants to consume the soft drink, he needs to impose some external force by his mouth. As we know capillary action takes place without the assistance of any external force, thus statement 2 is incorrect.

15. For which one of the following capillarity is not the only reason?

- (a) Blotting of ink
- (b) Rising of underground water
- (c) Spread of water drop on a cotton cloth
- (d) Rising of water from the roots of a plant to its foliage

I.A.S. (Pre) 1999

Ans. (d)

We can find several examples of capillarity in our daily life, such as blotting soft ink by blotting paper, rising of underground water, the spread of water drop on a cotton cloth, rising of kerosene in the wick of the lamp. But capillarity is not the only reason for rising of water from the roots of a plant to its foliage, apart from this transpiration and cohesion is also responsible for this process.

16. Kerosene rises in the wick of the stove is due to–

- (a) Osmosis
- (b) diffusion
- (c) Surface tension
- (d) biogen contraction

R.A.S./R.T.S. (Pre) 1997

Ans. (c)

Kerosene oil rising in the wick of the stove is due to the surface tension of oil. The wick of the lamp has many holes which act as capillaries. So kerosene keeps on rising in the capillaries.

17. In a capillary tube a fluid rises more as compared to water, it is due to :

- (a) Fluid is more viscous than that of water
- (b) Temperature of fluid is more than that of water
- (c) Surface tension of fluid is less than that of water
- (d) Surface tension of fluid is more than that of water

R.A.S./R.T.S.(Pre) 2007

Ans. (d)

In a capillary tube, liquid rises in proportion with its surface tension. The liquid which has more surface tension would gain more height.

18. When an air bubble at the bottom of lake rises to the top, it will–

- (a) Increased in size
- (b) Decreased in size
- (c) Maintain its size
- (d) Flatten into a disc like shape

I.A.S. (pre) 1996

Ans. (a)

When an air bubble at the bottom of the lake rises to the top, it will increase in size. The pressure of the trapped air in the bubble decreases due to decreasing vertical liquid depth and decrease in pressure exerted by the water.

19. While travelling by plane, the ink of the pen starts to come out–

- (a) Due to decrease in air pressure
- (b) Due to increase in air pressure
- (c) Due to increase in volume of ink
- (d) Due to excessive load.

U.P.P.C.S. (Pre) 1992

Ans. (a)

The fountain pen leak at high altitudes. It is because of low air pressure. The ink inside the fountain pen has high pressure compared to the low atmospheric pressure outside at high altitude, so ink flows out of the pen.

20. In automobile hydraulic brakes work on –

- (a) Archimedes Principle
- (b) Newton's law of motion
- (c) Bernoulli's Principle
- (d) Pascal's Law

Jharkhand P.C.S. (Pre) 2003

Ans. (d)

Hydraulic brake, hydraulic press, hydraulic elevator all works on the principle of Pascal's law which states that pressure exerted anywhere in a confined incompressible fluid is transmitted equally in all directions throughout the fluid such that the pressure variations remain the same.

21. A sudden fall in barometer reading indicates that the weather will be :

- (a) Stormy weather
- (b) Rainy weather
- (c) Cool dry weather
- (d) Hot and humid weather

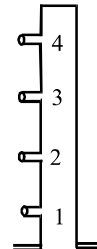
I.A.S. (Pre) 2001

U.P.P.C.S. (Pre) 1996

Ans. (a)

A sudden fall in barometer reading indicates that the weather will be stormy because atmospheric pressure decreases rapidly at the onset of a storm.

22. Consider the figure of a fountain with four holes, given below :



Which one of the holes in the fountain will throw the water farthest ?

- (a) 4
- (b) 3
- (c) 2
- (d) 1

I.A.S. (Pre) 1994

Ans. (d)

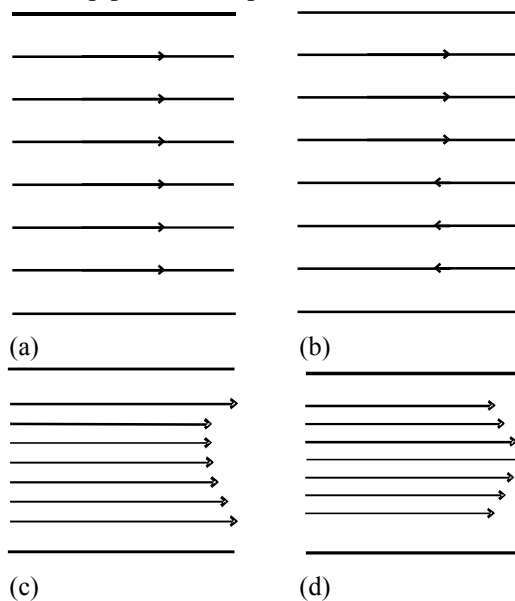
According to Torricelli's law,

$$\text{velocity of efflux} = \sqrt{2gh}$$

where, h is the depth of the hole.

In this question, hole 1 has maximum depth and it has maximum velocity of efflux, thus it will throw the water farthest.

23. A liquid is flowing in a streamlined manner through a cylindrical pipe. Along with a section containing the axis of the pipe, the flow profile will be :



Ans. (d)

U.P.P.C.S. (Pre) 2003

In a streamlined motion, the speed of layers of liquid is lowest near the pipe. As moved away from the pipe, the motion of layers gradually increases.

24. With the rise of temperature, the viscosity of liquid :

- (a) Increases
- (b) Decreases
- (c) Remains unchanged
- (d) May increase or decrease depending on nature of liquid

Uttarakhand P.C.S. (Pre) 2016

Ans. (b)

Viscosity is a measure of the resistance of a fluid which is being deformed by either shear stress or tensile stress. In everyday terms, viscosity is "thickness" or "internal friction". Viscosity of liquid (water) at different temperatures up to the normal boiling point is listed below.

Temperature (°C)	Viscosity (mPa.s)
10	1.308
20	1.002
30	0.7978
40	0.6531
50	0.5471
60	0.4668
70	0.4044
80	0.3550
90	0.3150
100	0.2822

Thus it is clear that with the rise of temperature, the viscosity of liquid decreases.

25. The highest viscosity among the following is of :

- (a) water
- (b) air
- (c) blood
- (d) honey
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

Viscosity is the measure of resistance of a fluid to flow. A fluid that is highly viscous has a high resistance (like having more friction) and flows slower than a low-viscosity fluid. Among the given options, honey has the highest viscosity.

26. Which one of the following is most viscous?

- (a) Alcohol
- (b) Water
- (c) Honey
- (d) Gasoline

Uttarakhand Lower Sub. (Pre) 2010

Ans. (c)

'Viscosity' of a fluid is a measure of its resistance to gradual deformation by shear stress or tensile stress. For liquids, it corresponds to the informal concept of 'Thickness'. Among the given options, honey is the most viscous fluid.

27. Assertion (A): With the increase of temperature, the viscosity of glycerin increases.

Reason (R) : Rise of temperature increases kinetic energy of molecules.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2002

Ans. (d)

Viscosity of any liquid (including glycerin) tends to decrease as its temperature increases. The increase in temperature causes the kinetic energy of molecules to increase and the molecules become more movable. Hence, (A) is false, but (R) is true.

28. Increasing the amount of a substance does not change its—

- (a) Volume
- (b) Weight
- (c) Mass
- (d) Density

U.P.P.C.S. (pre) 1992

Ans. (d)

Increasing the amount of substance does not change its density. Density of an object depends on its mass and volume. Density can be found by using the equation :

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

29. The surface of a lake is frozen in severe winter, but the water at its bottom is still liquid. Why?

Following are some options—

- (a) Ice is a bad conductor of heat
- (b) Since the surface of the lake is at the same temperature as the air, no heat is lost
- (c) The density of water is maximum at 4°C
- (d) None of the above

I.A.S. (Pre) 2011

Ans. (c)

Generally, all liquids expand on heating, but water is an exception to this rule. If water is heated from 0°C to 4°C, its volume gradually decreases. At temperatures over 4°C water starts expanding. It then keeps expanding with the further rise in temperature, till finally at 100°C and turns into steam. In other words, at 4°C, water has the least volume (occupies the least amount of space) and maximum density

(is at its heaviest). This irregular expansion of water is called anomalous expansion. This anomalous expansion plays an important role by only freezing the upper layer in lakes and rivers in severe winter. During winter months in colder countries, the outside or atmospheric temperature is very low – it drops to below freezing – and the upper layers of water in the lakes and ponds start cooling. When the temperature of the surface layers falls to 4°C, the water body acquires maximum density and sinks down. The water that sinks down displaces water below, and the lower layers of water simultaneously rise up. This also gets cooled to 4°C and again sinks down. When the temperature of the water body finally goes below 4°C, the density or heaviness of water decreases and as a result water does not sink down. The surface water finally freezes at 0°C while the lower part still remains at 4°C. The light frozen layer of ice floats on top.

30. In cold weather, aquatic animals survive even when water at the top layer of the lake freezes into ice because –

- (a) They can breathe in ice.
- (b) They have enough of accumulated oxygen inside them
- (c) Their body structure is such that they can survive without oxygen.
- (d) Water has the highest density at 4°C so underneath the top layer of ice there is a layer of water.

U.P.P.C.S. (Pre) 2005

Ans. (d)

This is because of a phenomenon is known as winter stratification in water bodies. In the winter season, the water at the top layer of water bodies will be ice cold (0°C) and are gradually frozen. But still, the bottom layers will have liquid water and maintain a temperature (4°C) which supports the life there. Thus, fishes and other aquatic animals are saved from the frozen top layer of water.

31. Fish can survive in a frozen lake because –

- (a) Fish are warm blooded animals
- (b) Fish hibernate in ice
- (c) Water near the bottom does not freeze
- (d) Ice is a good conductor of heat

Jharkhand P.C.S. (Pre) 2010

U.P.P.S.C. (GIC) 2010

Ans. (c)

See the explanation of above question.

32. When water is heated from 0°C to 10°C. It's volume :

- (a) Increases
- (b) Decreases
- (c) Does not change
- (d) First decreases and then increases

I.A.S. (Pre) 2001

Ans. (d)

An increase in temperature (heating) increases the volume of most liquids and consequently decreases the liquids density. But the behaviour of water from 0°C to 4°C is just opposite. When water is heated from 0°C to 4°C, then there is a decrease in volume and increase in density but when we cross the limit of 4°C then again it behaves like other normal liquids. Which means its volume increases and density decreases. Thus, when water is heated from 0°C to 10°C, it's volume first decreases and then increases.

33. What changes will the volume of water show if the temperature is lowered from 9°C to 3°C :

- (a) The volume will not change
- (b) The volume will first increase and then decrease
- (c) The volume will first decrease and then increase
- (d) Water will freeze

U.P.P.C.S. (Pre) 1997

Ans. (c)

If the temperature of water is lowered from 9°C to 4°C, then density is maximum at 4°C. Density is inversely proportional to volume, therefore till 4°C density increases which result in decrease in volume and later (4°C to 3°C) reduction in density would cause an increase in volume.

34. The density of water will reach its maximum value at the temperature of –

- (a) 0° C
- (b) 4° C
- (c) 50° C
- (d) 100° C

R.A.S./R.T.S. (Pre) 1992

Ans. (b)

The mass of any substance while heating remains the same however its volume increases. Thus, its density decreases. But water exhibits an anomalous expansion on cooling and contraction on heating within a specific range of temperature. When water at 0°C is heated, it is observed that it contracts between 0°C to 4°C, i.e. volume decreases and thus density increases. The volume of water is minimum at 4°C. Hence, the density of water is maximum at 4°C.

35. At which temperature density of water is maximum ?

- (a) 4° C
- (b) 0° C
- (c) -4° C
- (d) -8° C

U.P.P.C.S. (Mains) 2008

43rd B.P.S.C. (Pre) 1999

42nd B.P.S.C. (Pre) 1998

Ans. (a)

See the explanation of above question.

36. The maximum density of water is at :

- (a) 373 Kelvin
- (b) 277 Kelvin
- (c) 273 Kelvin
- (d) 269 Kelvin

U.P.P.C.S. (Mains) 2015

Ans. (b)

The density of water increases with temperature but volume decreases. At 4°C, the volume of water is low and density is maximum. Conversion of Celsius in Kelvin

$$K = C + 273$$

$$= 4 + 273 = 277$$

So, the density of water is maximum at 277 K.

37. What is the law in which under the same conditions of temperature and pressure, equal volumes of all gases contain equal number of molecules?

- (a) Coriolis effect (b) Graham's law
(c) Pascal's law (d) Avogadro's law
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

Avogadro's law states that under the same conditions of temperature and pressure, equal volumes of different gases contain an equal number of molecules. This empirical relation can be derived from the kinetic theory of gases under the assumption of a perfect (ideal) gas. For a given mass of an ideal gas, the volume and amount (moles) of the gas are directly proportional if the temperature and pressure are constant. The law is approximately valid for real gases at sufficiently low pressures and high temperatures.

38. Which of the following represent the suitable condition for the liquefaction of gases ?

- (a) Low temperature, high pressure
(b) Low temperature, low pressure
(c) High temperature, high pressure
(d) High temperature, low pressure

70th B.P.S.C. (Pre) 2024

Ans. (a)

Liquefaction is the process of turning a gas into a liquid. It's also known as condensation. Liquefaction is a complex process that involves many compressions and expansions to achieve the required high pressures and low temperatures. Increasing pressure for compressing a gas forces its molecules closer together. Decreasing a gas's temperature reduces the kinetic energy of its molecules. The molecules move more slowly and are more likely to be attracted to each other, forming a liquid.

39. In which of the following conditions distance between molecules of carbon dioxide filled in a container may increase?

- (i) On increasing the volume of container
(ii) On leakage of some carbon dioxide gas from the container
(iii) On increasing air pressure in the container
(iv) On adding more carbon dioxide gas in container without changing its size
(a) (ii) and (iii) (b) (iii) and (iv)
(c) (i) and (iii) (d) (i) and (ii)

Chhattisgarh P.C.S. (Pre) 2021

Ans. (d)

Gases do not have a definite volume and can occupy any amount of space available to them. Hence, on increasing the volume of container and on leakage of some carbon dioxide gas from the container, the distance between molecules of carbon dioxide filled in a container will increase. It will decrease on increasing air pressure or on adding more carbon dioxide in container.

40. If a gas is compressed to half of its original volume at 27°C, to what temperature should it be heated to make it occupy its original volume?

- (a) 327°C (b) 600°C
(c) 54°C (d) 300°C

I.A.S. (Pre) 1994

Ans. (a)

According to the law of gases

from formula $\frac{V_1}{V_2} = \frac{T_1}{T_2}$

$$V_1 = \frac{V}{2} \text{ and } V_2 = V \text{ [given]}$$

$$T_1 = 273 + 27 = 300 \text{ K}$$

$$T_2 = ?$$

$$\frac{V/2}{V} = \frac{300}{T_2} \Rightarrow T_2 = 600 \text{ K}$$

$$\therefore T_2 = 600 - 273 = 327^\circ\text{C}$$

41. Assertion (A) : An iron ball floats on mercury but gets immersed in water.

Reason (R) : The specific gravity of iron is more than that of mercury.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

I.A.S. (Pre) 2002

Ans. (c)

Specific gravity (relative density) is the measure of the density of an object in comparison to the density of water. An iron ball floats on mercury but gets immersed in water because the relative density of mercury is greater than the relative density of iron. However iron has a higher density than water, so it gets immersed in water. Thus assertion (A) is correct but reason (R) is incorrect.

42. Why does an iron nail float on mercury while it sinks in water ?

- (a) Due to less chemical interaction of iron with mercury than it is with water
(b) Iron is heavier than water and lighter than mercury.

- (c) Iron has greater density than water and lesser than mercury.
(d) Mercury is heavier than water.

U.P.P.C.S. (Pre) 1994

Ans. (c)

See the explanation of above question.

43. A needle sinks in water whereas a ship made of iron floats on it because :

- (a) The edge of the needle is pointed
(b) The ship is flat
(c) The ship drives by a powerful engine
(d) Relative density of the needle is greater than that of water displaced by it

U.P. Lower Sub. (Spl) (Pre) 2004

Ans. (d)

Relative density is defined as a ratio of the density of a particular substance with that of water. The relative density of pure water is 1. The substance having the relative density less than 1 will float in water, while substance having the relative density more than 1 will sink in water. The ship is designed in such a way that its effective density (m/v) is less than the density of water and the weight of the water displaced by the ship is equal to its weight, so it floats upon the water. But the density of iron needle is much larger than the water, so it sinks easily.

44. An iron needle floats on the surface of water due to –

- (a) Upthrust of water (b) Surface tension
(c) Viscosity (d) Gravitational force

U.P.P.C.S. (Mains) 2012

Ans. (b)

An iron needle can float on water due to the surface tension of water because upward force on the needle due to surface tension balances the weight of the needle.

45. While lifting a bucket of water from a well, we feel that the bucket :

- (a) Becomes heavier above the surface of water
(b) Becomes lighter above the surface of water
(c) Loses stability on coming out of the water
(d) Has gained mass on coming out of water

U.P.P.C.S. (Pre) 2009

Ans. (a)

According to the Archimedes' principle of flotation, object appears to be heavier in the air than inside the water or they appear to be less heavy in the water. It is because water exerts an upward force on the objects immersed in it. This reduction is equal to the weight of the fluid that body displaces.

46. 200 gm of water is filled in a weightless balloon. Its weight in water will be

- (a) Zero (b) 100 gm

(c) 200 gm

(d) 400 gm

U.P.R.O./A.R.O (Mains) 2014

Ans. (a)

Weight of balloon in water = Weight – Upthrust
= 200 – 200 = 0

The balloon is weightless.

47. The clouds float in the atmosphere because of their low :

- (a) Temperature (b) Velocity
(c) Pressure (d) Density

I.A.S. (Pre) 1995

Ans. (d)

Clouds form when the invisible water vapour in the air condenses into visible water droplets with very less density. As clouds have less density and high volume as compared to the air surrounding them, so the clouds experience a buoyant force from the surrounding air which balances their weight. Thus, the clouds float in the atmosphere because of their low density.

48. While floating in a sea what part of an iceberg is above the surface of the sea ?

- (a) 1/9 (b) 1/10
(c) 1/6 (d) 1/4

U.P.P.C.S. (Pre) 2005

Ans. (a)

When pure water changes into the ice at 4°C, the density of ice so formed is 9/10th of the density of water. Therefore, in pure water, 90% part of the ice must be below the surface of the water and remaining 10% part must be above the surface. Sea water has high density (salty) water but the ice formed by it is pure and not salty. Therefore, while floating in a sea about 1/8 to 1/9th part of an iceberg is above the surface of the sea.

49. An ice cube is floating in a glass of water. What happens to the water level when the ice melts? It –

- (a) Rises (b) Falls
(c) Remains the same (d) First rises and then falls

M.P.P.C.S. (Pre) 2008

Ans. (c)

According to Archimedes' principle, floating object displaces an amount of water equal to its own weight. Thus, the water level remains the same when the ice cube melts.

50. A piece of ice is floating in a beaker containing water up to its brim. When whole of the ice melts :

- (a) the water will spill on the floor
(b) the level of water will come down in the beaker
(c) the level of water will first fall and then it will spill out of the beaker